

Varun Tokas

Junior Undergraduate
Computer Science and Engineering, IIT Kanpur

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - 2025	B.Tech	Indian Institute of Technology Kanpur	9.72/10
2021	CBSE(XII)	Delhi Public School, RK Puram	96.4%
2019	CBSE(X)	Delhi Public School, RK Puram	95%

Scholastic Achievements

- Secured an **All India Rank** of **211** in **JEE Advanced - 2021** among the 2,50,000+ appearing candidates
- Secured an **All India Rank** of **840** in **JEE Mains - 2021** among the 10,00,000+ appearing candidates
- Received the **Academic Excellence Award** for exceptional academic performance in the 2021 and 2021-22 sessions
- Secured **All India Rank - 55** in **Kishore Vaigyanik Protsahan Yojana 2021** (Stream SX) among 1,50,000+ applicants
- Recipient of the **NTSE(National Talent Search Examination) 2019** scholarship for meritorious students in Class 10
- Placed in the **National Top 1%** of all students in the **Indian Olympiad Qualifier in Chemistry Part - I 2020-21**
- Placed in the **National Top 1%** of appearing students in the **Indian Olympiad Qualifier in Physics Part - I 2020-21**
- Placed in the **State Top 1%** of appearing students in the **Indian Olympiad Qualifier in Astronomy Part - I 2020-21**
- Awardee of the **Tower Research Capital Scholarship** (second place) in 2023 for second year undergraduate students

Project Experience

- CloudPhysician Vital Extraction Challenge | Inter IIT Tech Meet 11.0** (Jan'23-Feb'23)
 - Developed a **Computer Vision** pipeline to digitize information like Heart Rate, SpO₂ from images of ICU monitors
 - Used **semantic segmentation** to identify the monitor in an image and **warping** to remove perspective distortion
 - Tested various **OCR software** to determine feasible and reliable methods to extract information from the images
 - Applied image processing techniques to reduce the number of points in the polygonal bounding boxes predicted by semantic segmentation models like **YOLOv5-seg**, and approximate it using a **best-fit quadrilateral**
- CSE Bubble | Course Project (Computer Organization)** (Jan'23-Apr'23)
 - Designed a **32 bit** processor with **32 registers** and **25-instruction ISA** for sorting integer arrays using bubble sort
 - Implemented **instruction encoding and decoding modules** in **Verilog** and designed an ALU using a **top-down design approach** to reduce the hardware footprint of the processor and eliminate overheads during execution
 - Designed a **255 × 32 bit single port memory module** used as instruction and data memory for the processor
 - Implemented a **finite state machine** for **fetch-execute cycle** working in tandem with a program counter for fetching instructions from instruction memory, and wrote **testbenches** to carry out evaluation of the finished processor
- Neural Network from Scratch | Self Project** (Jul'23)
 - Implemented a neural network from scratch in Python using only **Numpy** to create **Dense** and **Activation** layers
 - Created a **Model** class with methods for training, adding layers, and specifying custom loss functions for training
 - Trained the network to learn **XOR** and also on the **MNIST** dataset, where it achieved a testing accuracy of **95%**

Competitive Programming

- Experienced competitive programmer with a peak rating of **1877 (Expert)** on **Codeforces** and **1796** on **Codechef**
- Global rank 33** in **Codechef Starters 24**(Division 3), and **Global rank 163** in **Codeforces Round 810**(Division 2)
- First place** in the Freshers' Programming Contest organized by Programming Club, IIT Kanpur out of 175 contestants

Technical Skills

- Programming Languages:** Python, C, C++, Verilog, MIPS
- Software and Libraries:** Pandas, Numpy, Matplotlib, Keras, TensorFlow, Bash, L^AT_EX, Scikit-Learn

Relevant Courses

((A*) indicates outstanding performance)

Data Structures and Algorithms (A*) Probability for Computer Science (A*) Fundamentals of Computing (A*) Software Development	Discrete Mathematics for Computer Science Computer Organization (A*) Linear Algebra and Differential Equations Logic for Computer Science
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Positions of Responsibility

- Secretary (Programming Club, IIT Kanpur)** (Jul'22 - Mar'23)
 - Responsible for organizing programming related **contests**, **lectures**, and **workshops** for the campus community
 - Wrote an introductory blog about **Domain Name System (DNS) requests**, **nameservers**, and **IP Addresses**
 - Organized, set problems for, and moderated the **Freshers' Programming Contest** for first-year students
- Student Guide (Counselling Service, IIT Kanpur)** (Oct'22 - Ongoing)
 - Responsible for helping and guiding 6 freshmen in adjusting emotionally and academically to the college environment
 - Coordinated with the Counselling Service and helped in the organization of **Orientation Programme 2022**